

Absolutely. I can create a **SQL Practice Handbook** designed specifically for hands-on learning, with **concepts** → **syntax** → **examples** → **exercises** → **solutions**, using the **W3Schools Northwind database** as the primary practice database.

A good structure would be:

SQL Practice Handbook

1. SQL Fundamentals

- SELECT
- DISTINCT
- WHERE
- AND / OR / NOT
- IN / BETWEEN
- LIKE / wildcards
- IS NULL

2. Sorting & Filtering

- ORDER BY
- ASC / DESC
- LIMIT / TOP
- Multiple-column sorting

3. SQL Functions

- COUNT, SUM, AVG, MIN, MAX
- String functions
- UPPER / LOWER
- CONCAT
- LEFT / RIGHT / SUBSTRING
- LENGTH
- Date functions
- Numeric functions
- Title Case

4. Grouping & Aggregation

- GROUP BY
- HAVING
- WHERE vs HAVING
- Aggregate functions

5. Joins

- INNER JOIN
- LEFT JOIN
- RIGHT JOIN
- FULL OUTER JOIN

- Self JOIN
- Multiple-table joins

6. Subqueries

- Scalar subqueries
- IN
- EXISTS
- Correlated subqueries

7. CASE Expressions

- Simple CASE
- Searched CASE
- Data classification

8. Set Operations

- UNION
- UNION ALL
- INTERSECT
- EXCEPT

9. Advanced SQL

- CTEs
- Window functions
- ROW_NUMBER
- RANK / DENSE_RANK
- LEAD / LAG

10. Data Modification

- INSERT
- UPDATE
- DELETE

11. DDL

- CREATE TABLE
- ALTER TABLE
- DROP TABLE
- Constraints
- Primary/Foreign Keys

12. SQL for Testers

- Backend validation
- UI vs DB validation
- API vs DB validation
- Test-data preparation
- Data integrity checks
- Duplicate detection
- Null validation

- Referential integrity

13. SQL Interview Practice

- Beginner exercises
- Intermediate exercises
- Advanced exercises
- Real-world QA scenarios
- Interview questions

14. Capstone Exercises

- Customer/order analysis
- Employee/sales analysis
- Product analysis
- Data-quality validation

I can make this as a **proper downloadable PDF handbook**, for example **100+ practical SQL exercises with answers**, using the W3Schools Northwind tables such as

Customers, Orders, OrderDetails, Products, Employees, Categories, and Suppliers.

I can also structure every exercise like:

Exercise 25 — Customer-wise Order Total

Requirement: Display customer name, order ID, product, quantity, price, and total amount.

Tables: Customers, Orders, OrderDetails, Products

Difficulty: ★★

Your Task: Write the SQL query.

Solution:

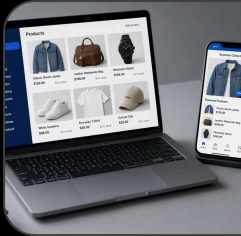
SQL

```
SELECT ...
```

Expected Result: sample output

Key Concept: INNER JOIN + calculated column

That format would work particularly well for **SQL training and QA/test-automation practice**.



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